

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – OCTOBER, 2006.

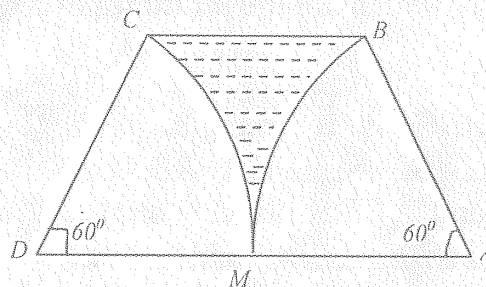
Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Attach the filled-in FACE SLIP and your rough working sheets along with the answer script.
5. Maximum time allowed is THREE hours.

1. If $a + \frac{1}{a} = -1$ find the value of $(a^2 + a + 1) + (a^4 + a^2 + 1) + (a^6 + a^3 + 1) + (a^8 + a^4 + 1) + \dots (a^{2006} + a^{1003} + 1)$

2. Given a system of equations $a+b = 2m^2$, $b+c = 6m$, $a+c = 2$, where m is a real number, find the range of real values of m such that $a \leq b \leq c$.

3. In quadrilateral $ABCD$, BC is parallel to AD . $\angle BAD = \angle CDA = 60^\circ$ circular arcs are drawn as shown in the diagram. M is the mid point of AD . If $AB = CD = \frac{1}{2}AD = 2$ units, determine the area of the shaded region.



4. If $\frac{1}{x+1} + \frac{2}{y+2} + \frac{2006}{z+2006} = 1$ prove that $\frac{x^2}{x^2+x} + \frac{y^2}{y^2+2y} + \frac{z^2}{z^2+2006z} = 2$.

5. A 2006 digit number has the property that every two consecutive digits as a two digit number is a multiple of 17 or 23. Find the maximum sum of the digits of such 2006 digit numbers.

6. Let p and q be positive integers such that $\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{54} + \frac{1}{55}$ prove that 83 divides p .

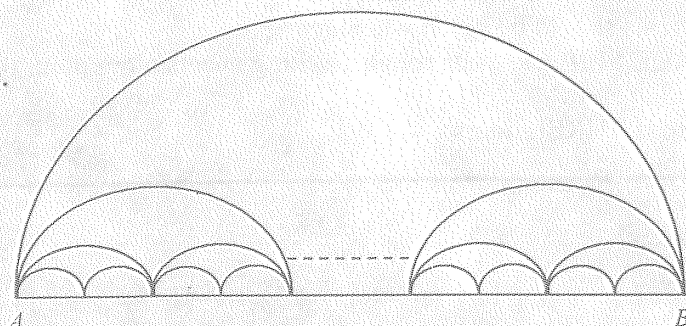
7. Find the sum of

$$\underbrace{2006^{100} + 2006^{100} + \dots + 2006^{100}}_{2006 \text{ times}} + \underbrace{2006^{101} + 2006^{101} + \dots + 2006^{101}}_{2005 \text{ times}} +$$

$$\underbrace{2006^{102} + 2006^{102} + \dots + 2006^{102}}_{2005 \text{ times}} + \dots + \underbrace{2006^{2005} + 2006^{2005} + \dots + 2006^{2005}}_{2005 \text{ times}}$$

8. Find the integral solution for x and y : $x^2 + 615 = 2^y$.

9.

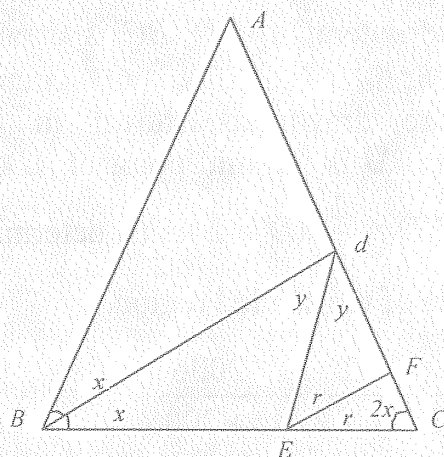


The number of smallest semicircles drawn on the diameter $AB=40.12mm$ is 2^n . The pattern as shown in the diagram is drawn. Show that the sum of the arc length of all the semicircular arcs is $(n+1) 20.06mm$.

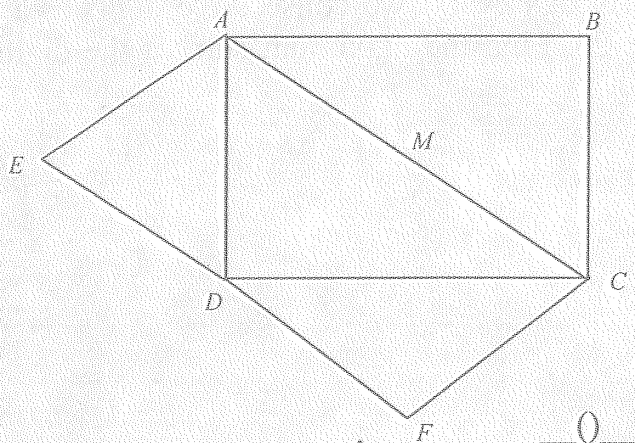
10. Find the largest number that cannot be expressed as a multiple 5, as a multiple of 8 or as the sum of multiple of 5 and a multiple of 8.

[Example: 15 is a multiple of 5, 24 is a multiple of 8, and 26 is the sum of 10 and 16 which are the multiples of 5 and 8.]

11. ABC is a triangle, with $\angle ABC = \angle ACB$, BD , DE and EF bisect angles $\angle ABE$, $\angle BDC$ and $\angle DEC$ respectively. $EF = EC$. Calculate the angles of the triangle ABC .



12.



$ABCD$ is a rectangle
 $AE=ED=DF=FC=\frac{1}{2}AC=AM$. Show
 that the sum of the areas of $\triangle AED$ and
 $\triangle DFC$ is equal to $\frac{1}{2}$ area of rectangle
 $ABCD$.